Jannik Zürn

Contact Information

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GitHub: https://github.com/jzuern

Education

Karlsruhe Institute of Technology, Karlsruhe, Germany

Master of Science, Theoretical Mechanical Engineering, Aug 2015 – Aug 2018
Graduated: Aug 2018
GPA: 3.7/4.0

- Thesis topic: *Neural Networks for Steady-State Fluid Flow Prediction*
  Advisors: Dr. S. Suwelack, Dr. Christof Megnin. Grade: 1.0

Bachelor of Science, Mechanical Engineering, Oct 2011 – Aug 2015
Graduated: Aug 2015
GPA: 3.1/4.0

- Thesis topic: *Numerical Solution of the Chemical Master Equation*
  Advisor: M.Sc. A. Koksharov. Grade: 1.0

Research Experience

Research Assistant June 2018 – Aug 2018
Renumics GmbH, Karlsruhe
Implementation of a novel data-driven approach to CFD into cloud computing frameworks

Research Assistant Sep 2016 – Apr 2018
FZI Research Center for Information Technology, Karlsruhe
Performance optimizations of TensorFlow, an open-source library for machine learning in various kinds of perceptual and language understanding tasks

Research Assistant Jan 2016 – Aug 2016
Institute for Biomedical Engineering,
Karlsruhe Institute of Technology
Continued development of a C++/MATLAB simulation of electrocardial signal transmission within the human heart

Research Assistant Oct 2014 – Apr 2015
Institute for Technical Thermodynamics,
Karlsruhe Institute of Technology
Implementation of numerical methods in C++ for the simulation of combustion processes

Research Assistant June 2013 – June 2014
Institute for Applied Computer Science,
Karlsruhe Institute of Technology
Development of a user interface in MATLAB and implementation of image processing algorithms for object tracking in live video

Publications

Rottmann, M., Zürn, J., Arslan, U., Klingel, K. and Dössel, O., 2016. *"Effects of fibrosis on the extracellular potential based on 3D reconstructions from histological sections of heart tissue."* Current Directions in Biomedical Engineering, 2(1), pp.675-678.

Working Experience

Summer Intern July 2017 – October 2017
Mayfield Robotics, Redwood City, California, USA
Supervisor: Kaijen Hsiao, Ph.D
Implementation of an artificial neural network for semantic recognition of a robot's environment with the goal of improving mapping and localization of a mobile robot

Summer Intern May 2015 – Sep 2015
ANSYS, Inc., San Diego, California, USA
Supervisor: Ellen Meeks, Ph.D
Conduction of case studies for the modeling and simulation of fluid mechanics and combustion chemistry and additionally development of software components in both C++ and Java

Extra-curricular Activities

- Personal software projects in C++ / Python, in the area of machine learning and numerical simulations — 2014 – present
- MOOCs: *Coursera: Machine Learning* (by Andrew Ng), *Coursera: Neural Networks for Machine Learning* (by Geoffrey Hinton)
- Participation at *talKIT*, Karlsruhe, the largest student-organized technology conference in Europe — May 2016
- Development of mobile application *Pocket Timer* — 2015
  Application-ID: *com.pockettimer*, available on Google Play Store
- Participation at music competition *Jugend Musiziert* as solo Cellist — 2008

Relevant Courses

Computational Intelligence, High Performance Computing, Computer Engineering, Numerical Methods and Simulation Techniques, Scientific Computing for Engineers, Continuum Mechanics, Robotics

Computer Skills

Linux Ubuntu, C/C++, MATLAB, Android SDK, Python, Fortran90, TensorFlow, LaTeX, git, ROS

Private Interests

Rock Climbing, Hiking, Cello

Language Skills

- German: *Mother tongue*
- English: *Business fluent*

References

Kaijen Hsiao, Ph.D
Chief Technology Officer
Mayfield Robotics — E-mail: kaijen@mayfieldrobotics.com

Ellen Meeks, Ph.D
Director of Development, Reacting Flow
ANSYS Inc. — E-mail: ellen.meeks@ansys.com